
DEMO

The Agentic Arm

Act. Check. Recover. One step at a time.

Franka Panda · PyBullet · Gemini Robotics-ER 2

one-shot vs agentic · 90 episodes · replay drift 0

Put every block in the bowl.

Three blocks. One bowl. A vision model driving the arm.

Plans every step up front, then executes blind.

One model call. It never reads a single result.

WHAT JUST HAPPENED

It left a block on the table.

Then it reported the job complete.

One action. Check the result. Decide again.

Same model. Same tools. Same budget.

WHAT JUST HAPPENED

It noticed, went back, and finished.

A placed block was removed. It put it back.

WHY BLIND PLANNING BREAKS

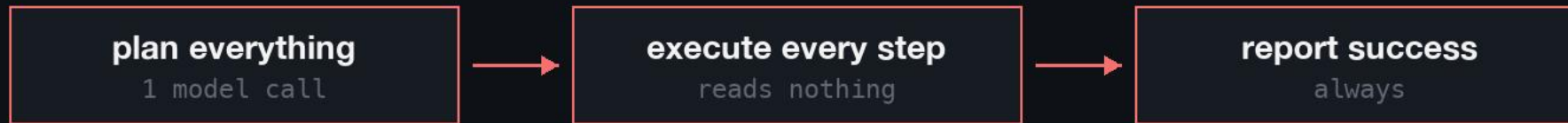
A plan written at $t=0$ cannot see $t=5$.

One early failure silently invalidates every step after it.

THE ONLY DIFFERENCE

One of them closes the loop

ONE-SHOT



AGENTIC



CHECKING, AFTER EVERY ACTION

Three checks, cheapest first

L1 Did it error?

the error string it already got back

free

L2 What do the fingers say?

0.000 m means it closed on empty air

free

L3 Does a photo agree?

one yes/no question about a fresh frame

1 call

The two free checks run first. A photo is only worth buying once they pass.

THE NUMBER THAT MATTERS

Robot A reported success in all 45 runs.

It finished half of them. Its report tracks nothing that happened.

THE RESULT

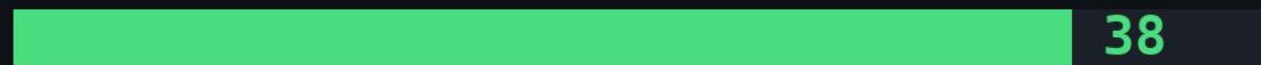
90 episodes, scored by ground truth

ROBOT A · one-shot



jobs actually finished, out of 45

ROBOT B · agentic



jobs actually finished, out of 45

51% → 84% same model, same tools, same budget

FALSE SUCCESSES

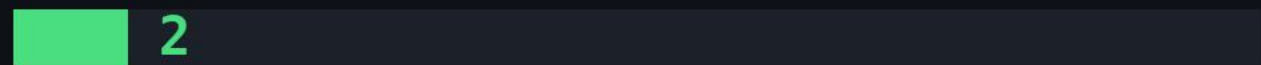
Claimed done. Ground truth disagreed.

ROBOT A · one-shot



episodes it claimed but had not done

ROBOT B · agentic



episodes it claimed but had not done

22 → 2 an operator told a job is done when it is not

EVERY TASK

Where the loop earned its keep

	ONE-SHOT	AGENTIC
1 block	5/5	5/5
2 blocks	5/5	5/5
3 blocks	4/5 1 false	5/5
match 3 colours	3/5 2 false	4/5 1 false
in a given order	4/5 1 false	5/5
swap two blocks	1/5 4 false	5/5
take one back out	1/5 4 false	0/5
3 blocks, one removed	0/5 5 false	5/5
match 3, one removed	0/5 5 false	4/5 1 false

THE HONEST LOSS

One task beat both of them.

Robot A claimed it 4 times anyway. Robot B, never once.

WHAT THE LOOP COSTS

Closing the loop is not free

1

model calls per run · one-shot

18

model calls per run · agentic

46

failures it caught and recovered from

All 45 live runs cost \$0.56 for Robot A, \$5.05 for Robot B.
Replaying them from the saved answers costs nothing.

GROUNDED IN A 2026 RESULT

VoLo · NVIDIA · arXiv:2606.07723

Its baseline family is literally "single action model,
no orchestrator". That is Robot A.

42.9% vs 14.3%

their orchestrated agent, on real hardware

84% vs 51%

ours, in a controlled miniature

We do not reproduce VoLo. That needs their benchmark and their robot.

THE TAKEAWAY

Judge an embodied agent on the honesty gap, not the leaderboard.

make judge

All 90 runs. Offline. No API key. \$0.00. Replay drift 0.

github.com/namanbansalcodes/agent-robot-arm